



FABRIC STRUCTURE & DESIGN

Part: B

Course Code: FE 305-0723

Compound Fabrics

Arnob Mahmud

Lecturer

Department of Fabric Engineering

Bangladesh University of Textiles (BUTEX)

E-mail: arnob@fe.butex.edu.bd

TUBULAR CLOTH

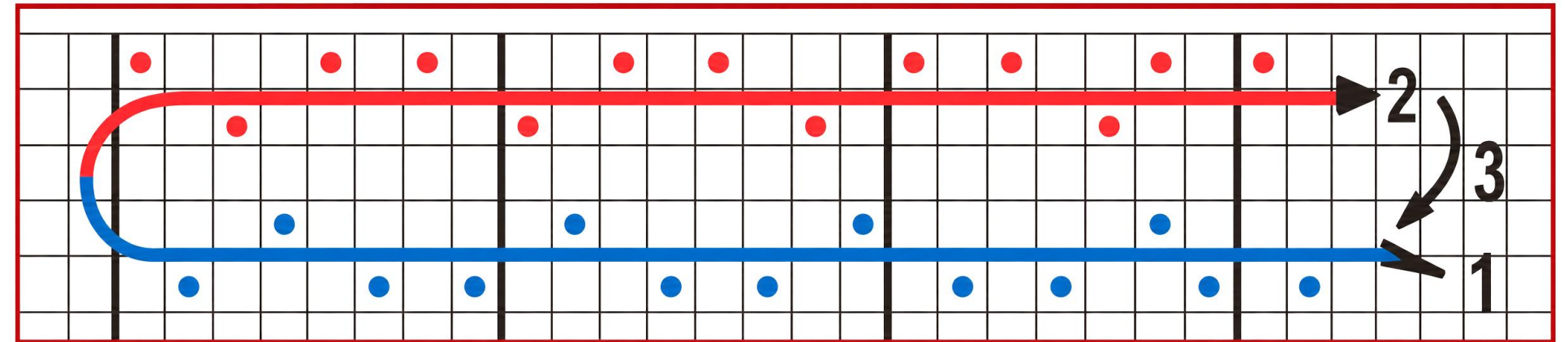
Tubular cloth is a type of double cloth *that forms a tube-like or hollow fabric*. It has two separate layers: a *face fabric* and a *back fabric*. *The two layers are joined at the selvages, so the fabric becomes tubular.*

- When the shuttle inserts the *face pick*, the *back warp threads are lowered*.
- When the shuttle inserts the *back pick*, the *face warp threads are raised*.
- In this way, face and back fabrics are *woven separately* but *joined at the edges*.
- Two sets of warp yarns are needed: *face warp and back warp*.
- Two sets of weft yarns are also needed: *face weft and back weft*.
- Usually, *the face and back warp may be wound on the same warp beam, and the same shuttle can insert both face and back picks*.

TUBULAR CLOTH

Application

- Fire hoses
- Seamless bags
- Sacks
- Technical drying cloths
- Decorative fabrics.



Typical Weaves

- Common base weaves are plain weave, 2/2 hopsack, 2/2 weft rib, 2/1 twill, and 2/2 twill.
- Plain weave is the most commonly used.

DOUBLE WIDTH CLOTH

Double width cloth is a fabric *woven as two layers*, but the *two layers are joined only at one side*. *The other side stays open.*

- It is made by *weaving 2 top picks and 2 bottom picks in sequence*.
- The top and bottom layers are *connected* by the weft at *only one edge*.
- The opposite edge remains open.
- Which side stays open depends on the order of weft insertion.
- *At the folded edge, the weft must pass continuously from one fabric layer to the other.*
- After weaving, the *fabric can be opened out* to get a *cloth about double the loom width*.

MULTI PLY FABRIC

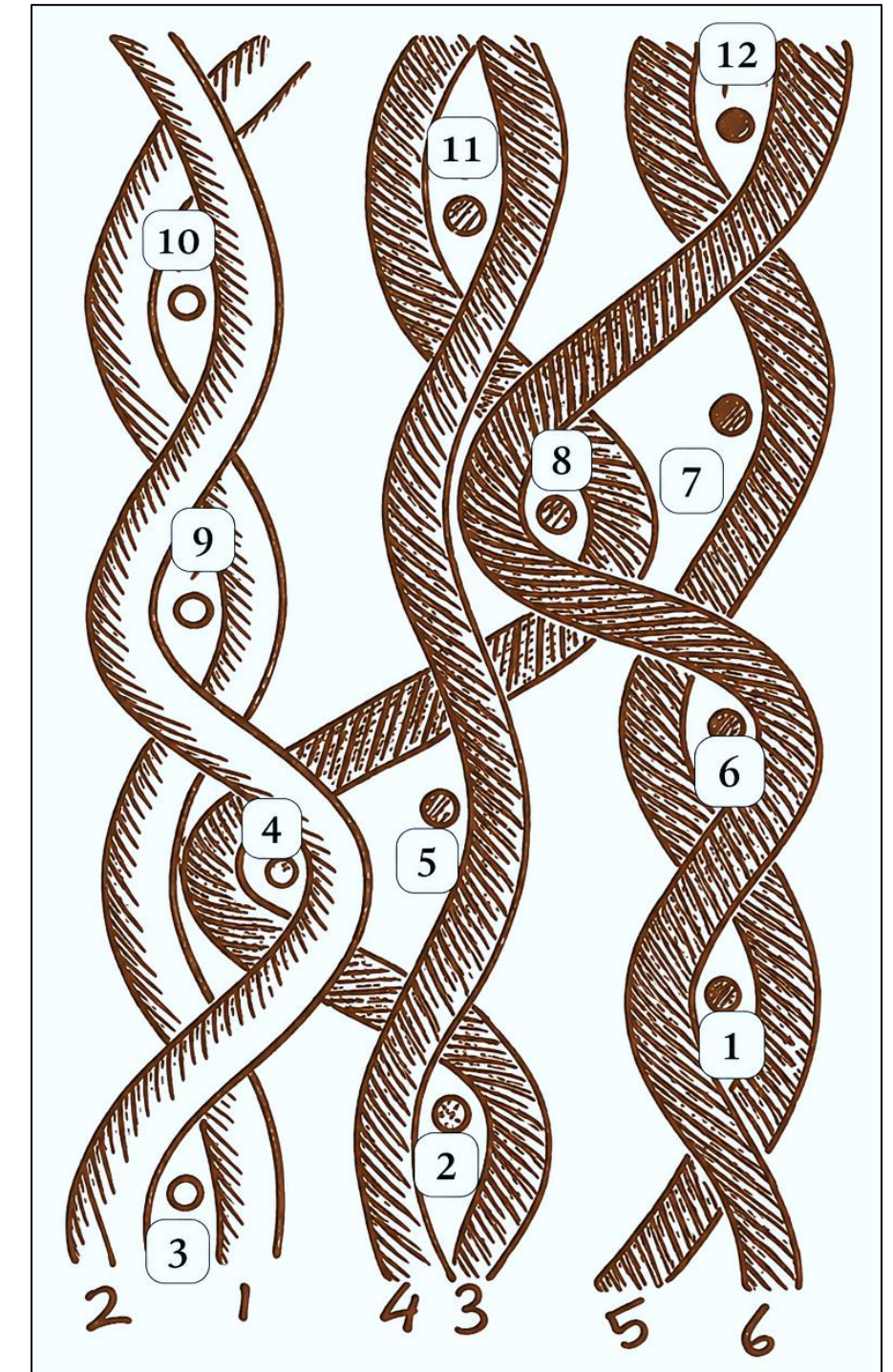
Basic Principle

- Multi-ply fabric is made of *three or more fabric layers*.
- The layers are *woven one above another*.
- Then the *layers are stitched or connected together*.
- Usually, 3 to 8 layers may be used.
- A narrow 8-ply fabric can be used for industrial belts.

Construction Principle

For a 3-ply fabric, there are three fabric layers:

- Face layer = Top layer
- Center layer = Middle layer
- Back layer = Bottom layer



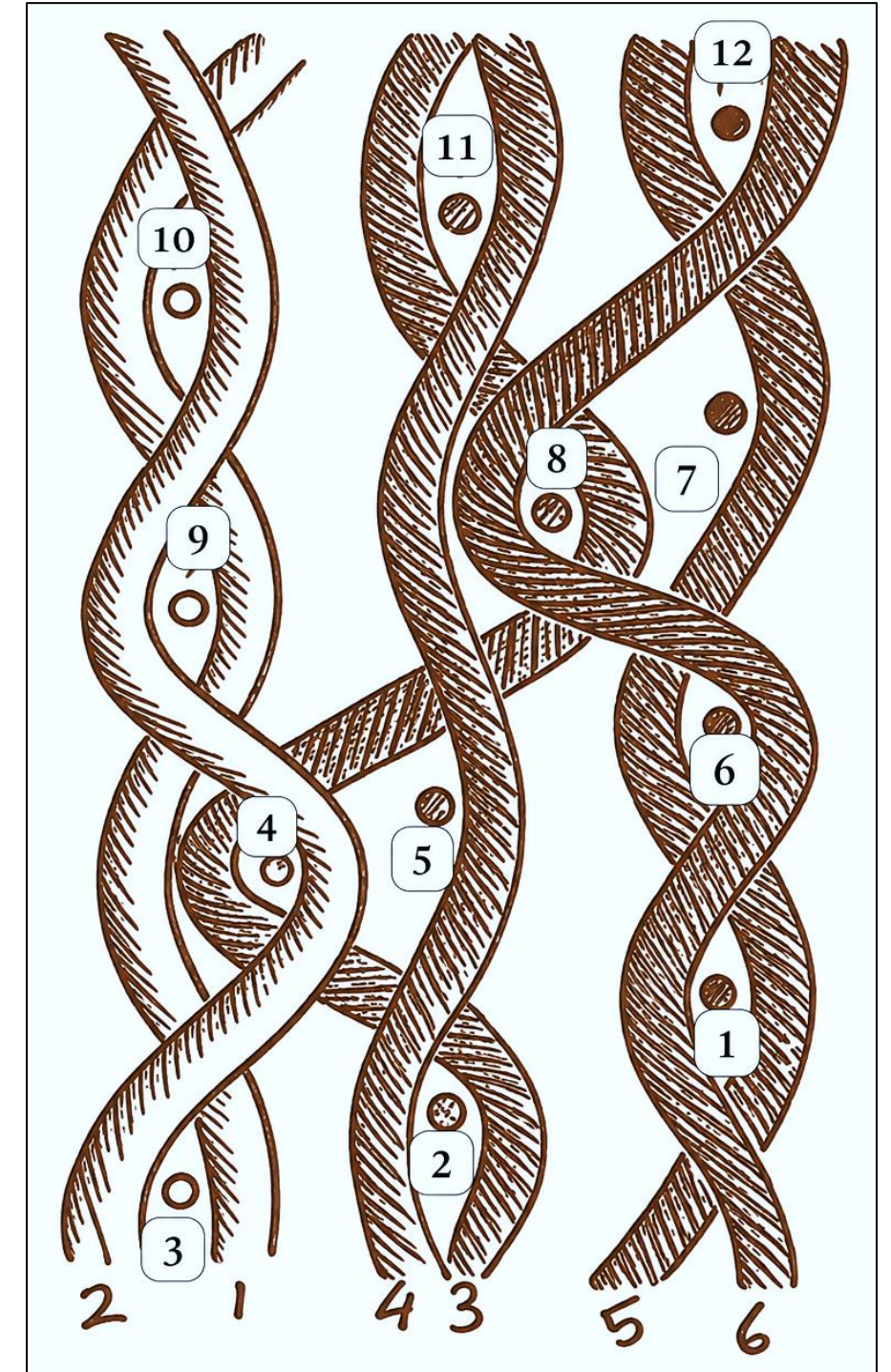
MULTI PLY FABRIC

Stitching Systems

The layers can be connected in different ways:

- ❖ Face → Center, then Center → Back
- ❖ Back → Center, then Center → Face
- ❖ Back → Center and Face → Center
- ❖ Center → Face and Center → Back
- ❖ Combination stitching

The main purpose is simply to join the separate fabric layers together.



MULTI PLY FABRIC

In the side Figure three layers are shown.

The stitching method used is:

Back → Center → Face

❖ Centre and Face joining

Centre warp thread 3 interlaces with face weft thread 4.

This joins the center fabric to the face fabric.

❖ Back and Centre joining

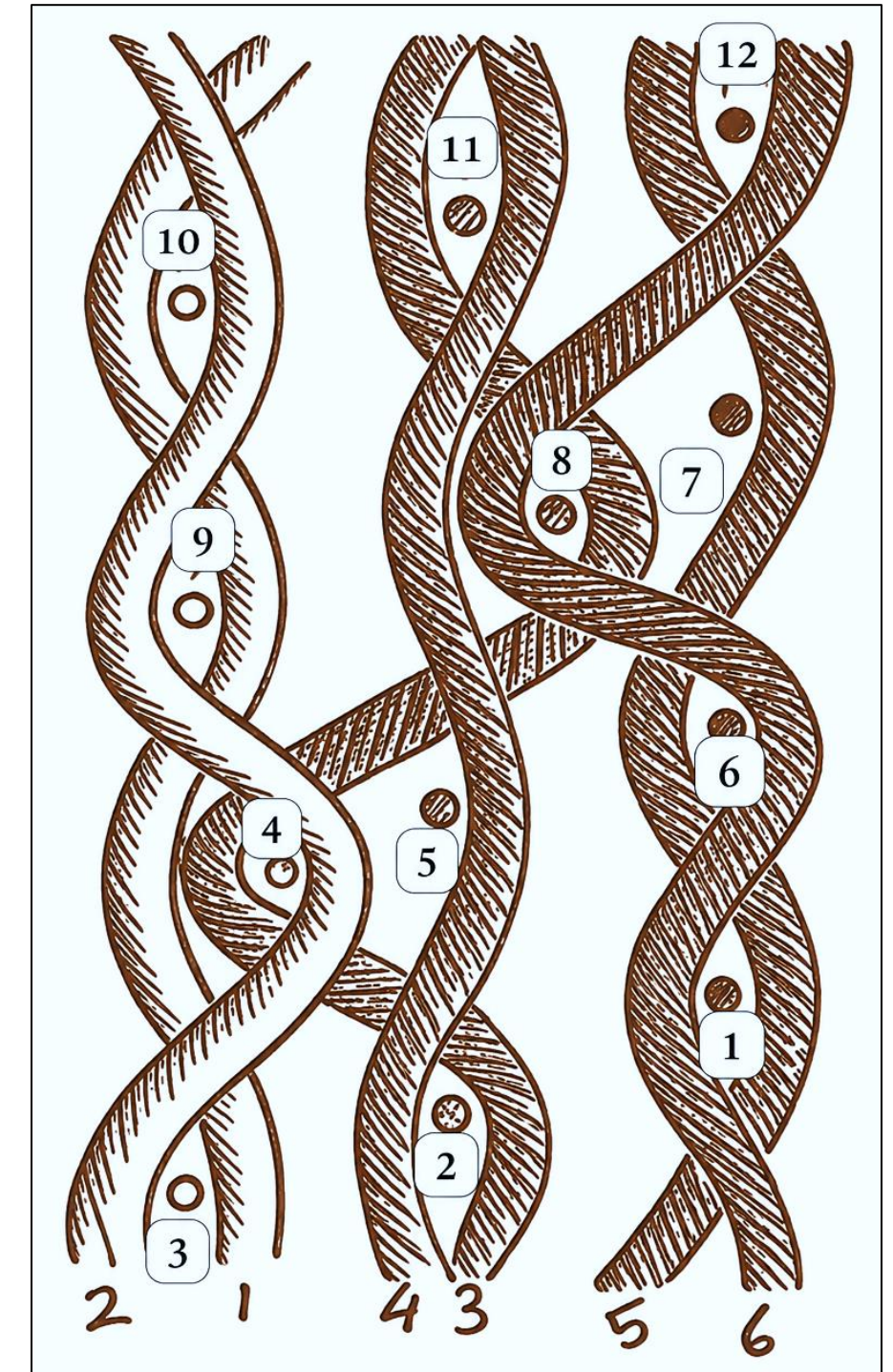
Back warp thread 6 interlaces with center weft thread 8.

This joins the back fabric to the center fabric.

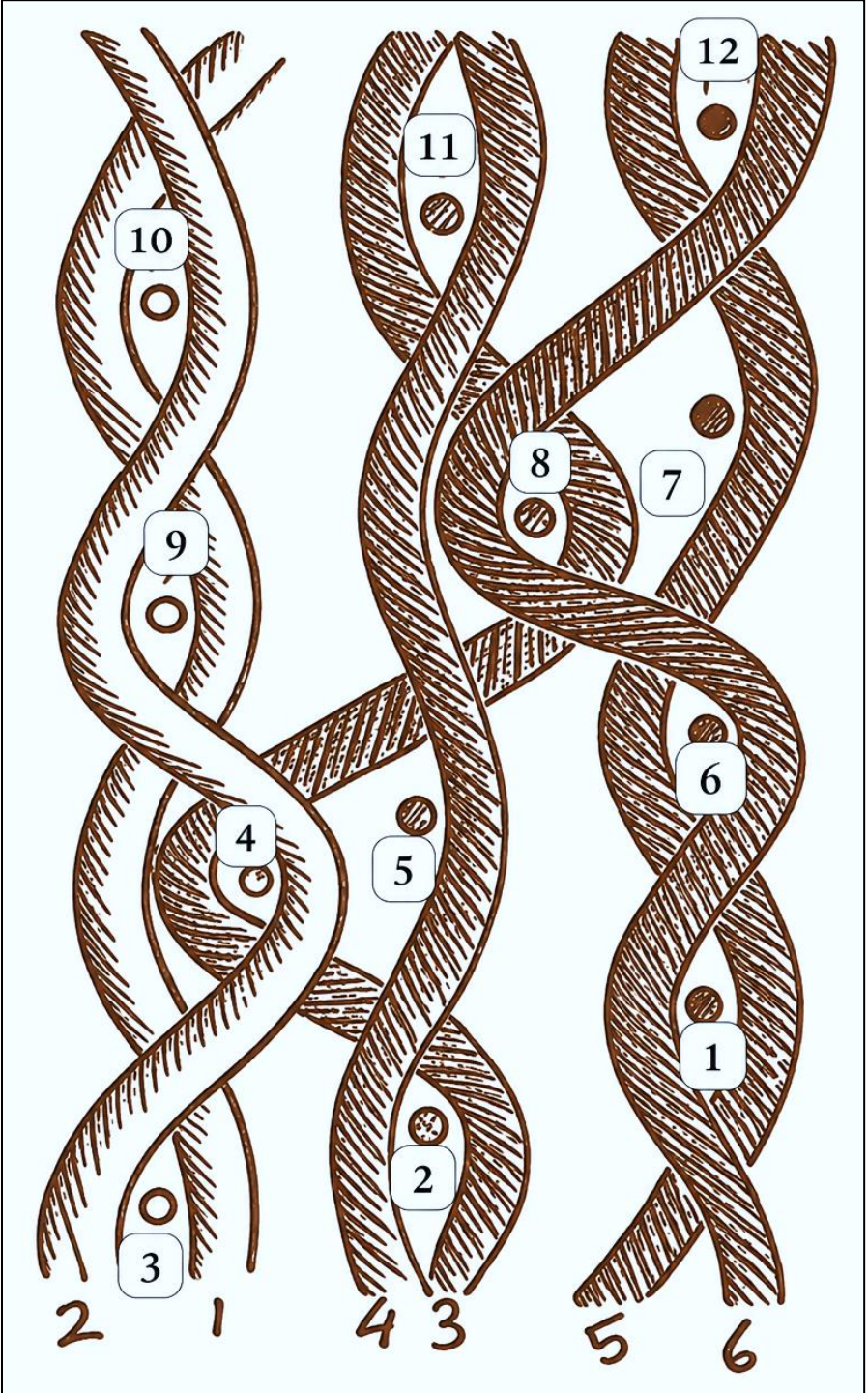
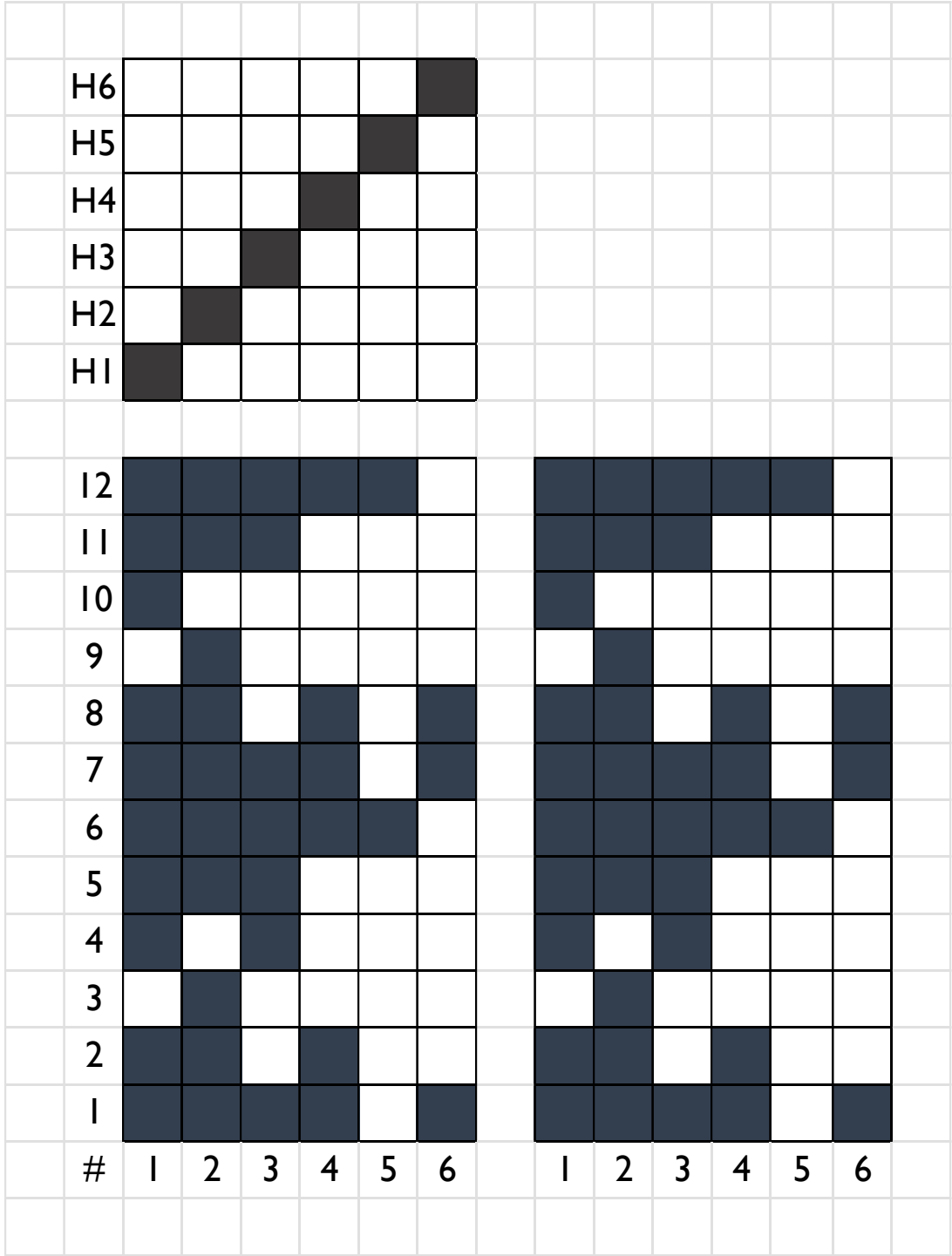
So finally:

Face layer ↔ Center layer ↔ Back layer

All three layers become connected.



MULTI PLY FABRIC



THANK
YOU !

